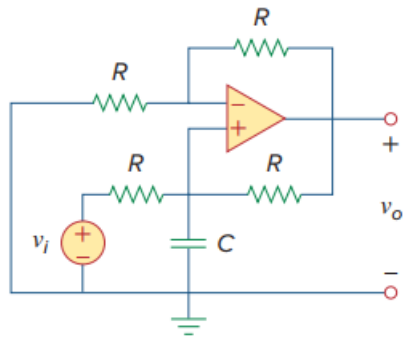


Problem

Show that the circuit in Fig. 6.90 is a noninverting integrator.

Fig. 6.90:



Step-by-step solution

Step 1 of 5

Refer to Figure 6.90 in the textbook.

Draw the circuit diagram with node notations.

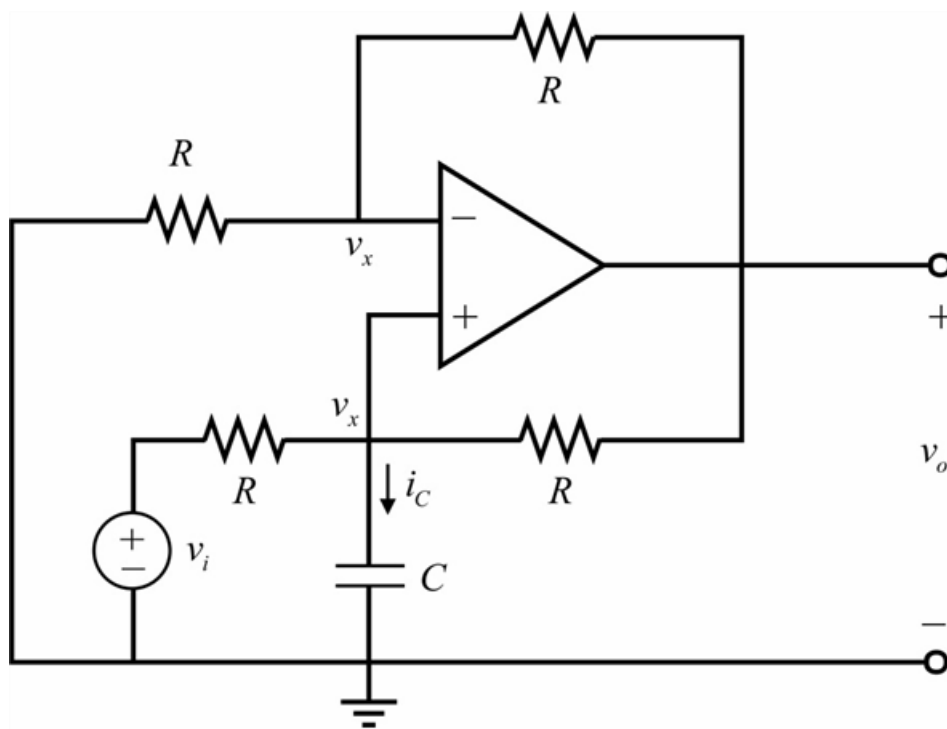


Figure 1

Step 2 of 5

As per the virtual ground concept, the voltage at the inverting terminal is equal to the voltage at the non-inverting terminal.

$$v^- = v^+ = v_x$$

Apply nodal analysis at the inverting terminal.

$$\frac{v_x}{R} + \frac{v_x - v_o}{R} = 0$$

$$v_x + v_x - v_o = 0$$

$$2v_x = v_o$$

$$v_x = \frac{v_o}{2}$$

Step 3 of 5

Consider the expression for current through the capacitor.

$$\begin{aligned} i_c &= C \frac{dv_c}{dt} \\ &= C \frac{dv_x}{dt} \end{aligned}$$

Apply nodal analysis at the non-inverting terminal.

$$\frac{v_x - v_i}{R} + \frac{v_x - v_o}{R} + i_c = 0$$

$$\frac{v_x - v_i + v_x - v_o}{R} + C \frac{dv_x}{dt} = 0$$

$$2v_x - v_i - v_o + RC \frac{dv_x}{dt} = 0$$

Step 4 of 5

Substitute $\frac{v_o}{2}$ for v_x .

$$2\left(\frac{v_o}{2}\right) - v_i - v_o + RC \frac{d}{dt}\left(\frac{v_o}{2}\right) = 0$$

$$v_o - v_i - v_o + \frac{RC}{2} \frac{dv_o}{dt} = 0$$

$$\frac{RC}{2} \frac{dv_o}{dt} = v_i$$

$$\frac{dv_o}{dt} = \left(\frac{2}{RC}\right) v_i$$

Step 5 of 5

Integrate on both sides.

$$\int \frac{dv_o}{dt} dt = \int \left(\frac{2}{RC}\right) v_i dt$$

$$\int dv_o = \frac{2}{RC} \int v_i dt$$

$$v_o = \frac{2}{RC} \int v_i dt$$

Thus, the output is proportional to the integral of input and output is positive.

Thus, the circuit is a noninverting integrator.